Calculation of I^2 in the Bayesian Framework

This supplementary file describes how I^2 was calculated for the Bayesian meta-analysis.

General definition

Higgins & Thompson (2002) define I^2 as the proportion of total variance in observed effect sizes that is attributable to between-study heterogeneity:

$$I^2 = \frac{\tau^2}{\tau^2 + v_t}$$

where τ^2 is the between-study variance and v_t is a measure of typical within-study variance across k included studies.

v_t is a weighted summary of all within-study sampling variances σ_k^2, computed by `metafor` (Viechtbauer 2010) as:

$$v_t = \frac{(k-1)\sum_k w_k}{\left(\sum_k w_k\right)^2 - \sum_k w_k^2}$$

where $w_k = 1/\sigma_k^2$ are the inverse-variance weights.

Bayesian calculation

As the formula above shows, v_t is calculated from within-study sampling variances σ_k^2 and therefore not specific to the Bayesian or Frequentist framework. We chose to reuse the value for v_t obtained from `metafor` instead of re-deriving v_t from σ_k^2, but both approaches would lead to the same result.

We calculated the distribution of I^2 by applying the I^2 formula from above to each posterior sample of τ^2.

Mean vs median

We summarised the posterior of τ^2 and I^2 using the median rather than the mean, because the median can be applied before or after the conversion from τ^2 to I^2 with the same result, whereas the mean cannot. In short, using the median avoids distorting the result, allowing us to compare the Bayesian to the Frequentist estimates.

References

Higgins, J. P., & Thompson, S. G. Quantifying heterogeneity in a meta-analysis. Statistics in medicine, 2002;21:1539–1558. https://doi.org/10.1002/sim.1186.

Wolfgang Viechtbauer. Conducting meta-analyses in with the metafor package. Journal of Statistical Software 2010;36:1–48. https://doi.org/10.18637/jss.v036.i03.